Experiment-5

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Course Code: -MLA0305 **Slot-**B

Title

Implementation of Dynamic Programming using Policy Iteration and Value Iteration

Aim

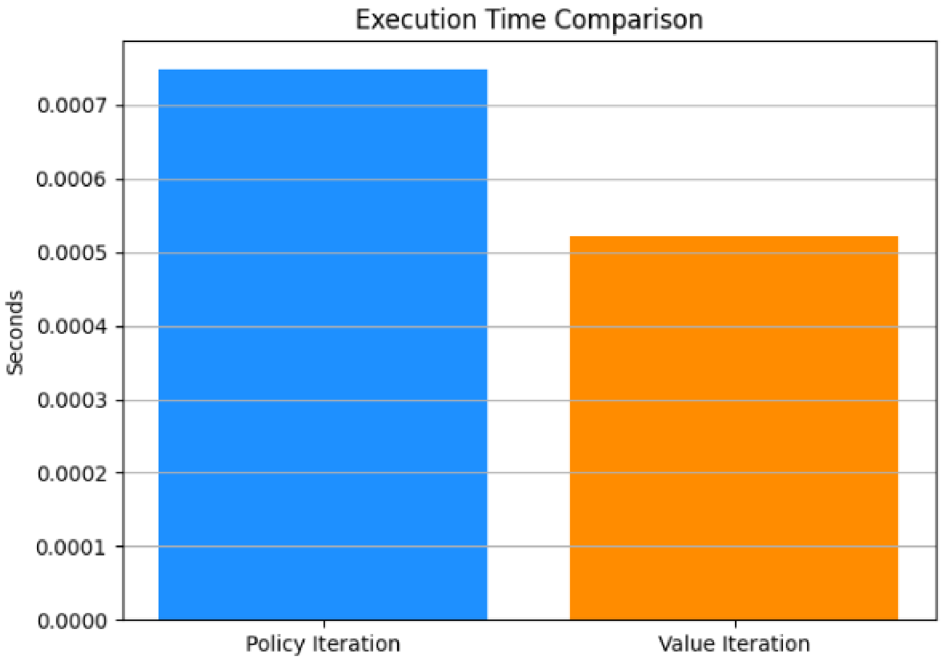
To implement **Dynamic Programming** methods, namely **Policy Iteration and Value Iteration**, for solving a Markov Decision Process and to determine the **optimal value function and optimal policy**.

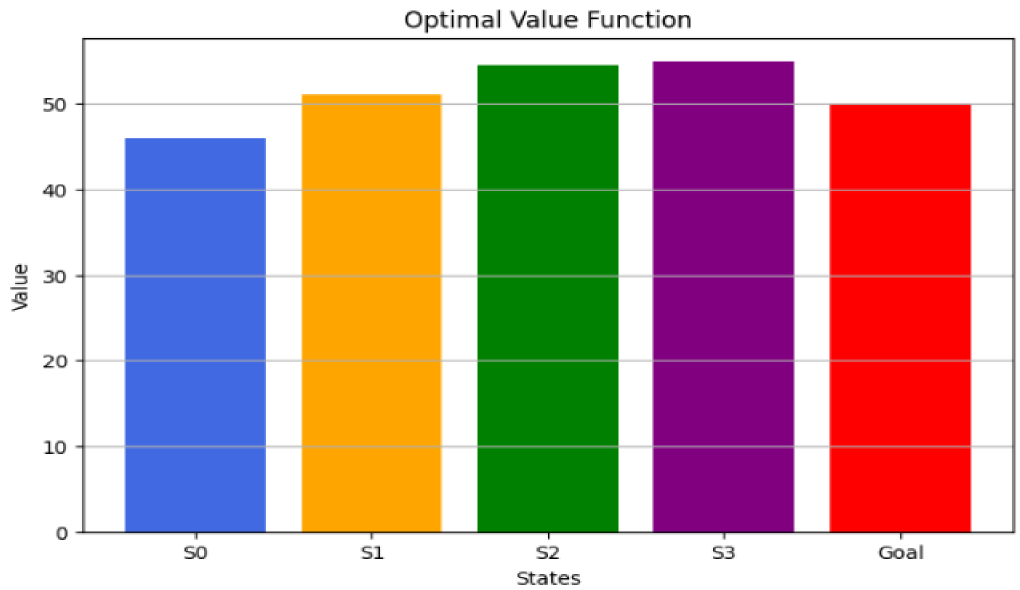
Procedure

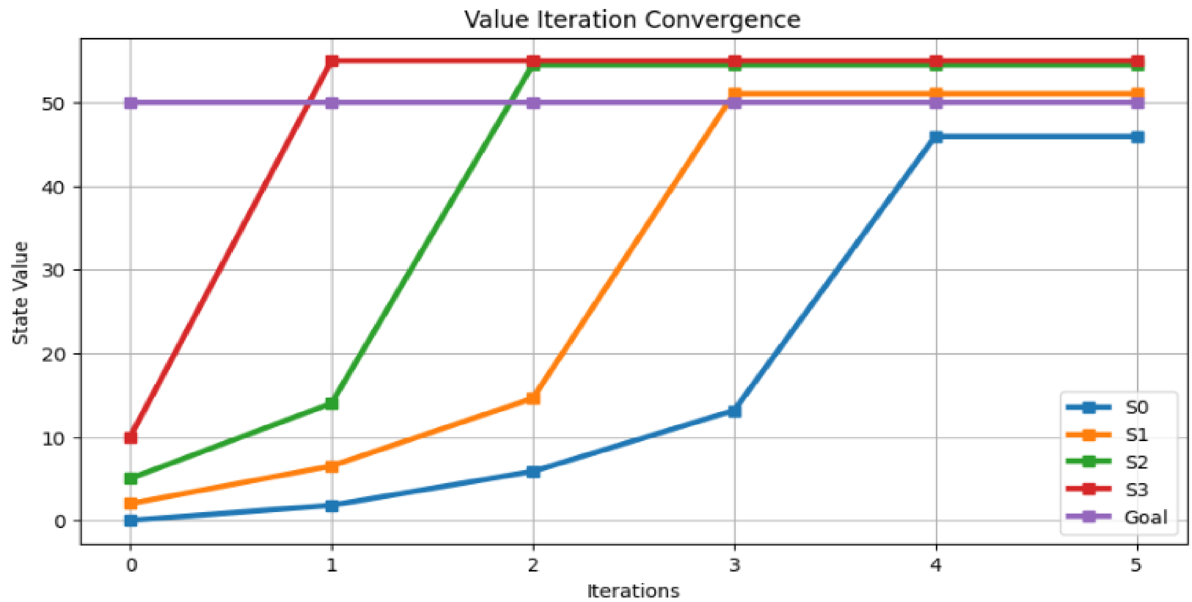
1. Define the MDP environment with a finite set of states and actions.
2. Define the transition function and reward associated with each state-action transition.
3. Select a discount factor .
4. Initialize the value function and an initial policy.
5. **Policy Iteration:**
   - Perform **policy evaluation** to calculate the value of each state under the current policy.
   - Perform **policy improvement** by selecting the action with the highest expected value.
   - Repeat evaluation and improvement until the policy becomes stable.
6. **Value Iteration:**
   - Initialize the value function.
   - Apply the Bellman optimality update:
   - $Vk + 1(s) = amax[R(s, a) + \gamma Vk(s')]$
   - Continue updating until the change in values is below a specified threshold.
7. Extract the optimal policy using:
   $$\pi * (s) = argamaxQ(s, a)$$
8. Compare Policy Iteration and Value Iteration based on iterations, execution time, state values, and optimal policy.
9. Generate convergence graphs and comparison graphs.

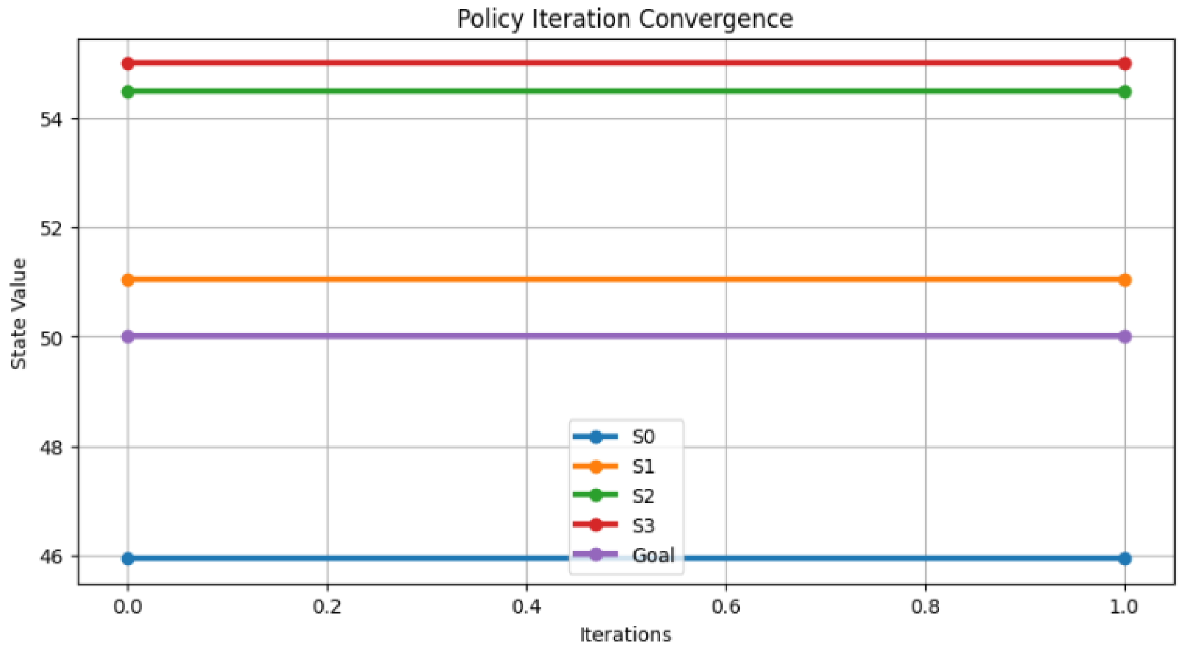
10. Display the final results using summary tables.
11. Export the results to CSV files.
12. Analyze which Dynamic Programming method performs better for the given MDP.

Visualizations




Optimal Value Function
Value
0
10
20
30
40
50
S0
S1
S2
S3
Goal
States


Value Iteration Convergence
State Value
0
10
20
30
40
50
0
1
2
3
4
5
Iterations
S0
S1
S2
S3
Goal


Policy Iteration Convergence
State Value
46
48
50
52
54
0.0
0.2
0.4
0.6
0.8
1.0
Iterations
S0
S1
S2
S3
Goal

Summary Table

State	Reward	Policy Iteration Value	Value Iteration Value	Optimal Policy
S0	0	Obtained from program	Obtained from program	Right
S1	2	Obtained from program	Obtained from program	Right
S2	5	Obtained from program	Obtained from program	Right
S3	10	Obtained from program	Obtained from program	Right
Goal	50	Obtained from program	Obtained from program	Terminal

Algorithm Comparison

Parameter	Policy Iteration	Value Iteration
Method	Policy Evaluation + Improvement	Bellman Optimality Updates
Policy	Improved iteratively	Extracted from optimal values
Value Function	(V^\pi(s))	(V^*(s))
Iterations	Obtained from program	Obtained from program
Execution Time	Obtained from program	Obtained from program
Optimal Policy	Obtained from program	Obtained from program

Result

Policy Iteration and Value Iteration were successfully implemented using Dynamic Programming. Both methods obtained the optimal value function and policy for the given MDP. Policy Iteration repeatedly performs policy evaluation and policy improvement, whereas Value Iteration directly applies the Bellman optimality equation to obtain optimal state values and subsequently extracts the optimal policy.